

Entropy: Process Discovery Experiment Report

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1 Introduction

This report summarizes the setup and results of a process discovery experiment conducted using the *Entropy* tool [?]. The following section outlines the discovery techniques used and their configurations in the experiment. Section 3 specifies the evaluation criteria used to assess the quality of the discovered models and to compare the performance of the discovery techniques. Finally, Section 4 presents the experimental results.

2 Discovery

The command line used to execute the experiment is listed below:

TODO: command line

Table 1 summarizes the key characteristics of the experimental setup used for performance comparison.

Table 1: Characteristics of the experiment

Parameter	Value
Population size	20
Iterations	25
Time limit (s)	2000
Pareto front size	320
Maximum model size	2000
DCI grid siz	10
Optimized models	SDAG
Entropic relevance background model	U
Event log	BPI_2013.xes

These discovery techniques are used in the experiment:

- DE [4];
- PSO [5];

3 Quality

The quality of the discovered models is assessed using two criteria: *entropic relevance*, which captures stochastic precision and recall [2], and *model size*, defined as the number of nodes and edges, which reflects model simplicity.

The performance of the discovery algorithms is evaluated using two aggregate measures: *dominance count* and the *Diversity Comparison Indicator* (DCI) [3]. Dominance count quantifies how many models discovered by a given algorithm dominate those from other algorithms in the global Pareto front. DCI measures the diversity of the discovered models by analyzing their distribution in the objective space. The details are provided in [1];

4 Results

Figure 1 presents the dominant discovered models with respect to entropic relevance and model size, where lower values are preferred for both criteria. Specifically, Fig. 1a, Fig. 1b, and Fig. 1c show the discovered deterministic frequency finite automata (DFFAs), stochastic deterministic finite automata (SDAGs), and directly-follows graphs (DFGs), respectively. The SDAG models are directly optimized, while the DFFA and DFG models are derived through translations of the optimized SDAGs. Table 5 in Appendix A lists the characteristics of all the discovered models.

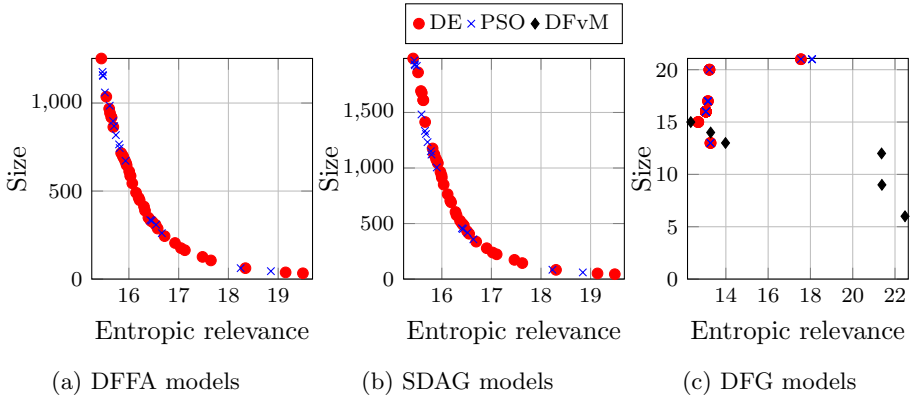


Figure 1: Discovered models plotted by size and entropic relevance for each representation type.

Table 2 and Table 3 summarize the analysis of the diversity of the models and the global Pareto front (dominance count), respectively. Table 4 reports the execution time (in iterations per second) for each input event log and method.

Table 2: Pareto dominance count comparison

Event log	DE(%)	PSO(%)	Total
BPI_2013.xes	77.78	22.22	9 100.00%

Table 3: DCI comparison (%)

Event log	DE(%)	PSO(%)
BPI_2013.xes	88.81	59.92

Table 4: Execution time comparison (iterations per second)

Event log	DE	PSO
BPI_2013.xes	513.05	1019.78

References

- [1] Hootan Zhian, Rajkumar Buyya, and Artem Polyvyanyy: *Multi-Objective Metaheuristics for Effective and Efficient Stochastic Process Discovery*. Proceedings of the 23rd International Conference on Business Process Management (BPM) Seville, Spain, August 31–September 5, 2025.
- [2] Hanan Alkhamash, Artem Polyvyanyy, Alistair Moffat, Luciano García-Bañuelos: *Entropic Relevance: A Mechanism for Measuring Stochastic Process Models Discovered From Event Data*. Inf. Syst. 107: 101922 (2022)
- [3] Miqing Li, Shengxiang Yang, Xiaohui Liu: *Diversity Comparison of Pareto Front Approximations in Many-Objective Optimization*. IEEE Trans. Cybern. 44(12): 2568–2584 (2014) Sander J.J. Leemans, Erik Poppe, Moe T. Wynn: *Directly Follows-Based Process Mining: Exploration & a Case Study*. In: ICPM, pp. 25–32 (2019)
- [4] Rainer Martin Storn, Kenneth Price: *Differential evolution - A simple and efficient adaptive scheme for global optimization over continuous spaces*. Tech. Rep. TR-95-012, International Computer Science Institute, 1947 Center Street, Berkeley (1995)
- [5] James Kennedy, Russell C. Eberhart: *Particle swarm optimization*. In: ICNN (1995)

A Model Characteristics

Table 5 summarizes the quality characteristics of discovered models produced by the selected algorithms.

Table 5: Model characteristics: size and entropic relevance

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFFA	Candidate1	x	330.0	16.449
DE	DFFA	Candidate2	x	349.0	16.392
DE	DFFA	Candidate3	x	491.0	16.144
DE	DFFA	Candidate4	x	692.0	15.888
DE	DFFA	Candidate5	x	968.0	15.604
DE	DFFA	Candidate6	x	389.0	16.324
DE	DFFA	Candidate7	x	412.0	16.304
DE	DFFA	Candidate8	x	612.0	16.006
DE	DFFA	Candidate9	x	647.0	15.955
DE	DFFA	Candidate10	x	658.0	15.945
DE	DFFA	Candidate11	x	544.0	16.069
DE	DFFA	Candidate12	x	1253.0	15.447
DE	DFFA	Candidate13	x	675.0	15.919
DE	DFFA	Candidate14	x	586.0	16.033
DE	DFFA	Candidate15	x	943.0	15.621
DE	DFFA	Candidate16	x	919.0	15.647
DE	DFFA	Candidate17	x	164.0	17.125
DE	DFFA	Candidate18	x	287.0	16.578
DE	DFFA	Candidate19	x	448.0	16.217
DE	DFFA	Candidate20	x	126.0	17.480
DE	DFFA	Candidate21	x	586.0	16.028
DE	DFFA	Candidate22	x	462.0	16.196
DE	DFFA	Candidate23	x	700.0	15.879
DE	DFFA	Candidate24	x	1036.0	15.546
DE	DFFA	Candidate25	x	106.0	17.648
DE	DFFA	Candidate26	x	864.0	15.687
DE	DFFA	Candidate27	x	919.0	15.652
DE	DFFA	Candidate28	x	33.0	19.497
DE	DFFA	Candidate29	x	38.0	19.149
DE	DFFA	Candidate30	x	62.0	18.342
DE	DFFA	Candidate31	x	176.0	17.046
DE	DFFA	Candidate32	x	205.0	16.927
DE	DFFA	Candidate33	x	308.0	16.537
DE	DFFA	Candidate34	x	244.0	16.718
DE	DFFA	Candidate35	x	335.0	16.430
DE	DFFA	Candidate36	x	325.0	16.473
DE	DFFA	Candidate37	x	716.0	15.845
PSO	DFFA	Candidate38	x	45.0	18.854
PSO	DFFA	Candidate39	x	309.0	16.547
PSO	DFFA	Candidate40	x	672.0	15.933
PSO	DFFA	Candidate41	x	818.0	15.738
PSO	DFFA	Candidate42	x	985.0	15.615
PSO	DFFA	Candidate43	x	1159.0	15.477

Method	Type	Name	Visualization	Size	Entropic relevance
PSO	DFFA	Candidate44	x	1060.0	15.518
PSO	DFFA	Candidate45	x	261.0	16.662
PSO	DFFA	Candidate46	x	62.0	18.246
PSO	DFFA	Candidate47	x	332.0	16.457
PSO	DFFA	Candidate48	x	869.0	15.703
PSO	DFFA	Candidate49	x	1154.0	15.482
PSO	DFFA	Candidate50	x	744.0	15.816
PSO	DFFA	Candidate51	x	335.0	16.430
PSO	DFFA	Candidate52	x	765.0	15.803
PSO	DFFA	Candidate53	x	897.0	15.683
PSO	DFFA	Candidate54	x	1177.0	15.470
DE	SDAG	Candidate1	x	492.0	16.425
DE	SDAG	Candidate2	x	525.0	16.367
DE	SDAG	Candidate3	x	764.0	16.117
DE	SDAG	Candidate4	x	1118.0	15.851
DE	SDAG	Candidate5	x	1693.0	15.580
DE	SDAG	Candidate6	x	575.0	16.297
DE	SDAG	Candidate7	x	604.0	16.276
DE	SDAG	Candidate8	x	965.0	15.972
DE	SDAG	Candidate9	x	1043.0	15.920
DE	SDAG	Candidate10	x	1056.0	15.909
DE	SDAG	Candidate11	x	851.0	16.037
DE	SDAG	Candidate12	x	1986.0	15.422
DE	SDAG	Candidate13	x	1081.0	15.883
DE	SDAG	Candidate14	x	913.0	16.000
DE	SDAG	Candidate15	x	1677.0	15.597
DE	SDAG	Candidate16	x	1612.0	15.623
DE	SDAG	Candidate17	x	223.0	17.106
DE	SDAG	Candidate18	x	408.0	16.551
DE	SDAG	Candidate19	x	691.0	16.187
DE	SDAG	Candidate20	x	173.0	17.465
DE	SDAG	Candidate21	x	934.0	15.994
DE	SDAG	Candidate22	x	705.0	16.170
DE	SDAG	Candidate23	x	1126.0	15.842
DE	SDAG	Candidate24	x	1861.0	15.521
DE	SDAG	Candidate25	x	144.0	17.623
DE	SDAG	Candidate26	x	1414.0	15.664
DE	SDAG	Candidate27	x	1610.0	15.628
DE	SDAG	Candidate28	x	44.0	19.488
DE	SDAG	Candidate29	x	51.0	19.140
DE	SDAG	Candidate30	x	83.0	18.303
DE	SDAG	Candidate31	x	239.0	17.025
DE	SDAG	Candidate32	x	278.0	16.905
DE	SDAG	Candidate33	x	431.0	16.509

Method	Type	Name	Visualization	Size	Entropic relevance
DE	SDAG	Candidate34	x	337.0	16.693
DE	SDAG	Candidate35	x	501.0	16.406
DE	SDAG	Candidate36	x	482.0	16.444
DE	SDAG	Candidate37	x	1176.0	15.814
PSO	SDAG	Candidate38	x	60.0	18.843
PSO	SDAG	Candidate39	x	418.0	16.518
PSO	SDAG	Candidate40	x	1005.0	15.898
PSO	SDAG	Candidate41	x	1233.0	15.712
PSO	SDAG	Candidate42	x	1482.0	15.590
PSO	SDAG	Candidate43	x	1934.0	15.452
PSO	SDAG	Candidate44	x	1919.0	15.496
PSO	SDAG	Candidate45	x	358.0	16.636
PSO	SDAG	Candidate46	x	84.0	18.231
PSO	SDAG	Candidate47	x	451.0	16.428
PSO	SDAG	Candidate48	x	1306.0	15.680
PSO	SDAG	Candidate49	x	1925.0	15.457
PSO	SDAG	Candidate50	x	1121.0	15.792
PSO	SDAG	Candidate51	x	457.0	16.406
PSO	SDAG	Candidate52	x	1151.0	15.779
PSO	SDAG	Candidate53	x	1334.0	15.660
PSO	SDAG	Candidate54	x	1968.0	15.445
DE	DFG	Candidate1	x	16.0	13.054
DE	DFG	Candidate2	x	16.0	13.054
DE	DFG	Candidate3	x	16.0	13.054
DE	DFG	Candidate4	x	16.0	13.054
DE	DFG	Candidate5	x	20.0	13.222
DE	DFG	Candidate6	x	16.0	13.054
DE	DFG	Candidate7	x	16.0	13.054
DE	DFG	Candidate8	x	16.0	13.054
DE	DFG	Candidate9	x	16.0	13.054
DE	DFG	Candidate10	x	16.0	13.054
DE	DFG	Candidate11	x	16.0	13.054
DE	DFG	Candidate12	x	21.0	17.550
DE	DFG	Candidate13	x	16.0	13.054
DE	DFG	Candidate14	x	16.0	13.054
DE	DFG	Candidate15	x	21.0	17.550
DE	DFG	Candidate16	x	20.0	13.222
DE	DFG	Candidate17	x	15.0	12.693
DE	DFG	Candidate18	x	16.0	13.054
DE	DFG	Candidate19	x	16.0	13.054
DE	DFG	Candidate20	x	15.0	12.693
DE	DFG	Candidate21	x	16.0	13.054
DE	DFG	Candidate22	x	16.0	13.054
DE	DFG	Candidate23	x	16.0	13.054

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFG	Candidate24	x	20.0	13.222
DE	DFG	Candidate25	x	15.0	12.693
DE	DFG	Candidate26	x	17.0	13.153
DE	DFG	Candidate27	x	20.0	13.222
DE	DFG	Candidate28	x	13.0	13.273
DE	DFG	Candidate29	x	13.0	13.273
DE	DFG	Candidate30	x	13.0	13.273
DE	DFG	Candidate31	x	15.0	12.693
DE	DFG	Candidate32	x	16.0	13.054
DE	DFG	Candidate33	x	16.0	13.054
DE	DFG	Candidate34	x	16.0	13.054
DE	DFG	Candidate35	x	16.0	13.054
DE	DFG	Candidate36	x	16.0	13.054
DE	DFG	Candidate37	x	17.0	13.153
PSO	DFG	Candidate38	x	13.0	13.273
PSO	DFG	Candidate39	x	16.0	13.054
PSO	DFG	Candidate40	x	16.0	13.054
PSO	DFG	Candidate41	x	17.0	13.153
PSO	DFG	Candidate42	x	20.0	13.222
PSO	DFG	Candidate43	x	21.0	18.074
PSO	DFG	Candidate44	x	21.0	17.550
PSO	DFG	Candidate45	x	16.0	13.054
PSO	DFG	Candidate46	x	13.0	13.273
PSO	DFG	Candidate47	x	16.0	13.054
PSO	DFG	Candidate48	x	17.0	13.153
PSO	DFG	Candidate49	x	21.0	18.074
PSO	DFG	Candidate50	x	17.0	13.153
PSO	DFG	Candidate51	x	16.0	13.054
PSO	DFG	Candidate52	x	17.0	13.153
PSO	DFG	Candidate53	x	17.0	13.153
PSO	DFG	Candidate54	x	21.0	18.074